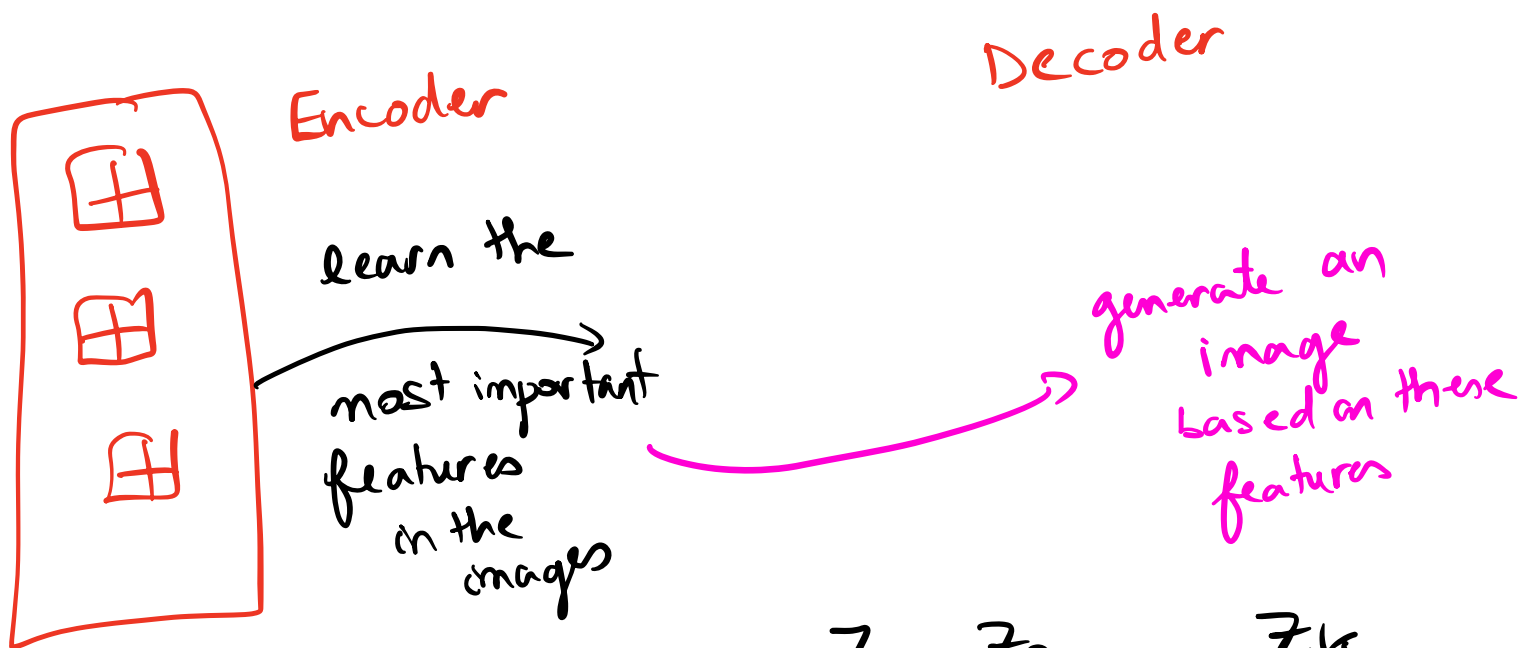


CS 6180 10/22

Review from last time

generation of images

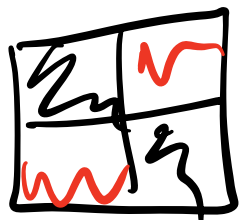
* Variational auto encoders



features: $z_1, z_2, \dots, z_k$
latent (not observed)

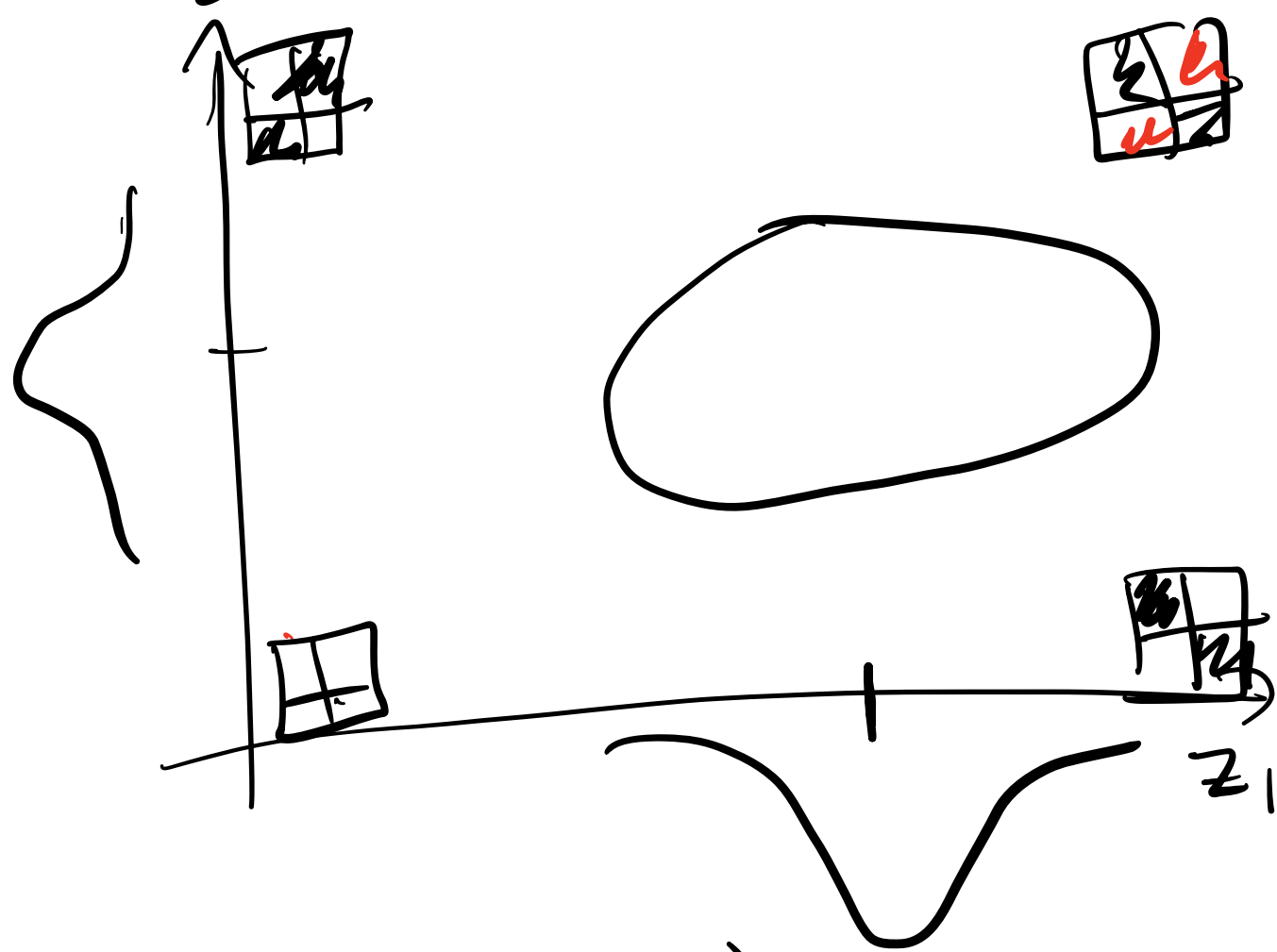
For simplicity, let's consider two features

z_1, z_2



z_2

latent space



$$p(x | z_1) = N(\mu_1, \sigma_1^2)$$

$$p(x | z_2) = N(\mu_2, \sigma_2^2)$$

Metric to compare probability distributions:

KL divergence

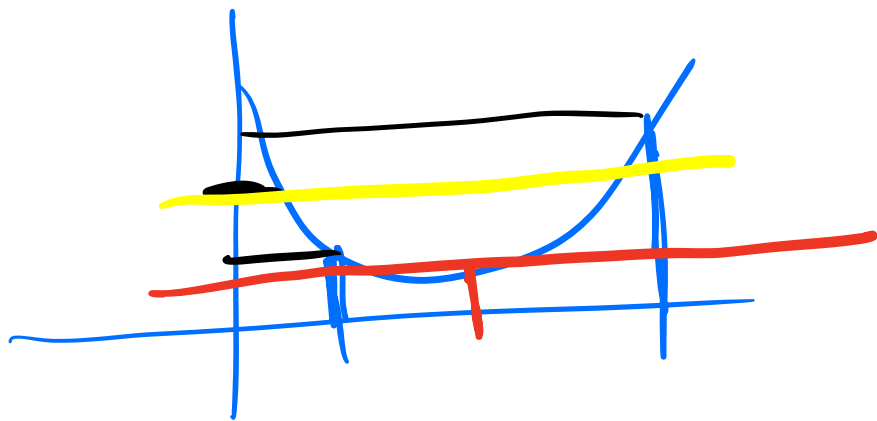
$$KL(p, q) = \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right)$$

* $KL(p, q) \geq 0$ (that requires Jensen's inequality)

Jensen's inequality

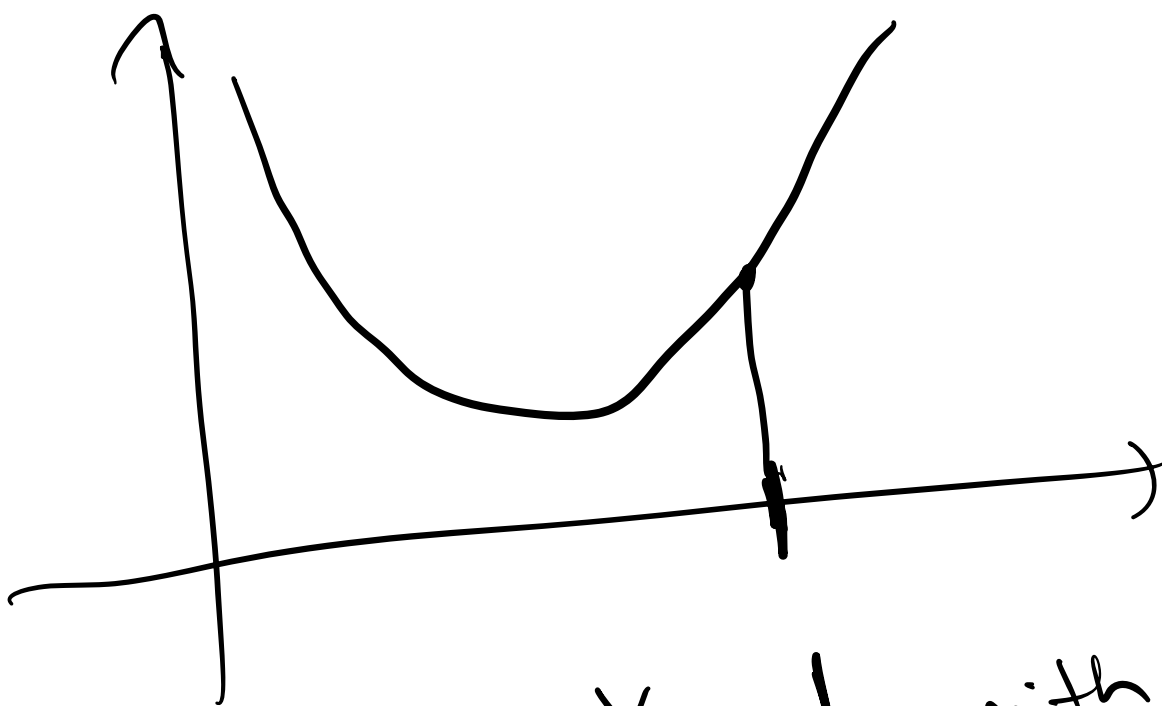
If f is a convex function and X is any random variable then

$$f(\underline{E(X)}) \leq E(f(X))$$



Jensen's inequality becomes an equality if $X = E(X)$

(i.e. X is just a constant)



$X = 1$ with proba 1
 $E(X) = 1$

$$\begin{aligned} \beta(E(x)) &= \beta(1) \\ &= \\ E(\beta(x)) &= E(\beta(1)) = \beta(1) \end{aligned}$$

We used Jensen's inequality

$$\begin{aligned} KL(p, q) &= \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right) \\ &= E_{p(x)} \left[\log \left(\frac{p(x)}{q(x)} \right) \right] \\ &= E_{p(x)} \left[-\log \left(\frac{q(x)}{p(x)} \right) \right] \end{aligned}$$

but $-\log$ is convex

$$E_{p(x)} \left[-\log \left(\frac{q(x)}{p(x)} \right) \right]$$

$$\begin{aligned}
& \sum_x -\log \left[E_p(x) \left(\frac{q(x)}{p(x)} \right) \right] \\
&= -\log \left[\sum_x p(x) \cdot \frac{q(x)}{p(x)} \right] \\
&= -\log \left[\sum_x q(x) \right] \\
&= 0.
\end{aligned}$$

$$KL(p, q) = 0 \Leftrightarrow p = q$$

$$\begin{aligned}
KL(p, q) &= E_{p(x)} \left[-\log \left(\frac{q(x)}{p(x)} \right) \right] \\
&= -\log \left[E_{p(x)} \left(\frac{q(x)}{p(x)} \right) \right] \\
&= 0
\end{aligned}$$

X

$$\text{if } \frac{q(x)}{p(x)} = c$$

$$\Rightarrow q(x) = c \cdot p(x)$$

$$\Rightarrow \underbrace{\sum_x q(x)}_1 = \sum_x c p(x) = c \cdot (1) = c$$

$$\Rightarrow c=1 \quad \Rightarrow q(x) = p(x)$$

$$* \text{KL}(p, q) \quad \text{vs} \quad \text{KL}(q, p)$$

$$\sum_x p(x) \log\left(\frac{p(x)}{q(x)}\right) \neq \sum_x q(x) \log\left(\frac{q(x)}{p(x)}\right)$$

antisymmetric

P_{data} , P_{θ}
↓
 P_{model}

$KL(P_{data}, P_{\theta})$ vs $KL(P_{\theta}, P_{data})$

P_{data} is the correct
and we are computing
the error I get if I
were to use

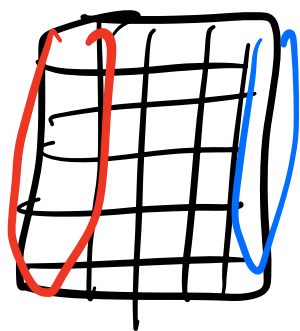
P_{θ}

loss
What? function to optimize?

Likelihood
of the data
occurring

|| Likelihood function

$\vec{x}^{(1)}, \vec{x}^{(2)}, \vec{x}^{(3)}, \dots, \vec{x}^{(n)}$



turn into a vector



$p(x^{(1)})$

high

$p(x^{(2)})$

high

...

$p(x^{(n)})$

$p(x^{(1)} \& x^{(2)} \& \dots \& x^{(n)})$ high

much more challenging

independence of sample to deal with

$$= p(x^{(1)}) \cdot p(x^{(2)}) \cdots p(x^{(n)})$$

$$= \prod_{i=1}^n p(x^{(i)}) \leftarrow \text{likelihood function}$$

$\Rightarrow$ we have to compute derivatives to maximize this quantity

derivatives of products are not fun (product rule)

log likelihood function

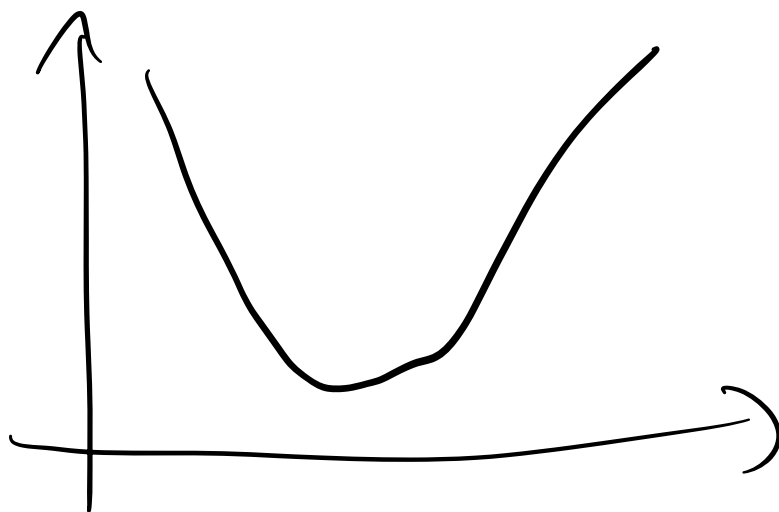
$$\log \left(\prod_{i=1}^n p(x^{(i)}) \right) = \sum_{i=1}^n \log p(x^{(i)})$$

let's just focus on one example X

$$\log p(x) = \log \left(\sum_z p(x|z) \cdot p(z) \right)$$

very hard to optimize
(not convex)

$$p(x) = \sum_z \underbrace{p(x|z)} \cdot p(\underline{z})$$



convex
optimization

Let's come up with a strategy to help us optimize the log likelihood function

$$\log p_{\theta}(x) = \log \left(\sum_z p(x, z) \right)$$

$$= \log \left(\sum_z \frac{p(x, z)}{Q(z)} Q(z) \right)$$

for some probability distribution
 $Q(z)$

$$= \log \left(\sum_z \frac{p(x, z)}{Q(z)} \cdot \underline{\underline{Q(z)}} \right)$$

$$= \log \left(E_{Q(z)} \left(\frac{p(x, z)}{Q(z)} \right) \right)$$

$$\geq E_{Q(z)} \left(\log \left(\frac{p(x, z)}{Q(z)} \right) \right)$$

ELBO
evidence lower bound

using
Jensen's
inequality

$$f(E(x)) \leq E(f(x)) \quad f \text{ is convex}$$

$$g(E(x)) \geq E(g(x)) \quad g \text{ is concave}$$

this will turn into an
equality if

$$\frac{P(x, z)}{Q(z)} = c$$

$$\Rightarrow Q(z) = \frac{P(x, z)}{c}$$

$$\Rightarrow \sum_z Q(z) = \sum_z \frac{p(x|z)}{c}$$

$$\Rightarrow 1 = \frac{1}{c} p(x)$$

$x_1 \backslash x_2$	1	2
1	$\frac{1}{4}$	$\frac{1}{8}$
2	$\frac{1}{2}$	$\frac{1}{8}$

$$\sum_{x_2} p(x_1=1, x_2) = p(x_1=1) \\ = \frac{1}{4} + \frac{1}{8}$$

$$\Rightarrow c = p(x)$$

$$\Rightarrow Q(z) = \frac{p(x|z)}{c}$$

$$\Rightarrow Q(z) = \frac{p(x|z)}{p(x)}$$

$$= \frac{p(x) \cdot p(z|x)}{p(x)}$$

$$\Rightarrow \boxed{Q(z) = p(z|x)}$$

$$\log p_{\theta}(x) = \mathbb{E}_{p(z|x)} \left[\log \left(\frac{p(x|z)}{p(z|x)} \right) \right]$$

$$= E_{p(z|x)} \left[\log \left(\frac{p(z) \cdot p(x|z)}{p(z|x)} \right) \right]$$

$$= E_{p(z|x)} \left[\log(p(x|z)) + \log \left(\frac{p(z)}{p(z|x)} \right) \right]$$

$$= E_{p(z|x)} [\log(p(x|z))] + E_{p(z|x)} \left[\log \frac{p(z)}{p(z|x)} \right]$$

$$= E_{p(z|x)} [\log p(x|z)] - E_{p(z|x)} \left[\log \frac{p(z|x)}{p(z)} \right]$$

$$= E_{p(z|x)} [\log p(x|z)] - \sum_z p(z|x) \log \left(\frac{p(z|x)}{p(z)} \right)$$

$$= \boxed{E_{p(z|x)} [\log p(x|z)] - \text{KL}(p(z|x), p(z))}$$

$\log p(x)$

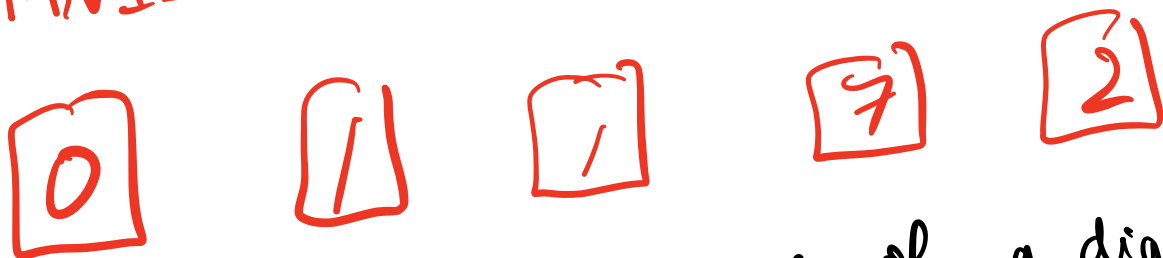
reconstruction
loss

KL divergence
term

hard to know what it is

Let's write the VAE model now:

MNIST dataset



Goal: generate an image of a digit

$$\vec{x}^{(1)}, \vec{x}^{(2)}, \dots, \vec{x}^{(n)}$$

n black/white images

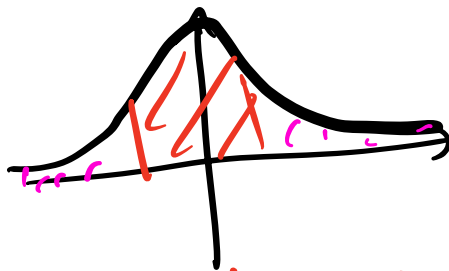
$$x_i \in \{0, 1\}$$

K latent variables

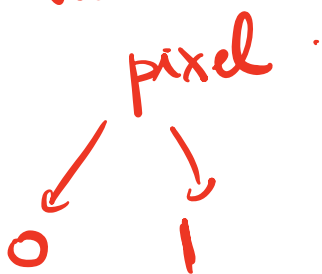
$$\vec{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_K \end{pmatrix}$$

* first generate some random z values

$$p(z_i) = N(0, 1) \text{ for } i=1, \dots, K$$



* Use the z values to generate a



$$p(x|z) = \text{Bernoulli}(\vec{p}(z))$$

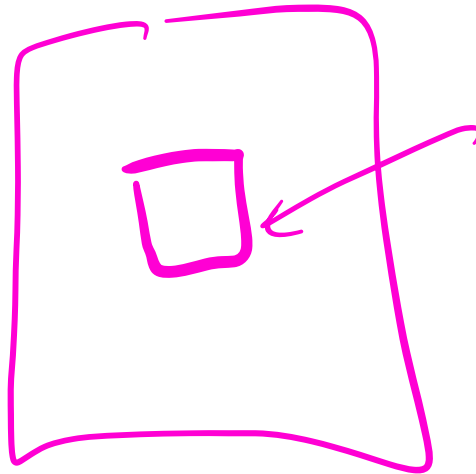
/ Bernoulli

$$p(\vec{x} | z) = \begin{cases} \text{Bernoulli} \\ \text{Bernoulli} \end{cases}$$

$z = 0.2$ → model noise

$z = 0.2$

model will learn



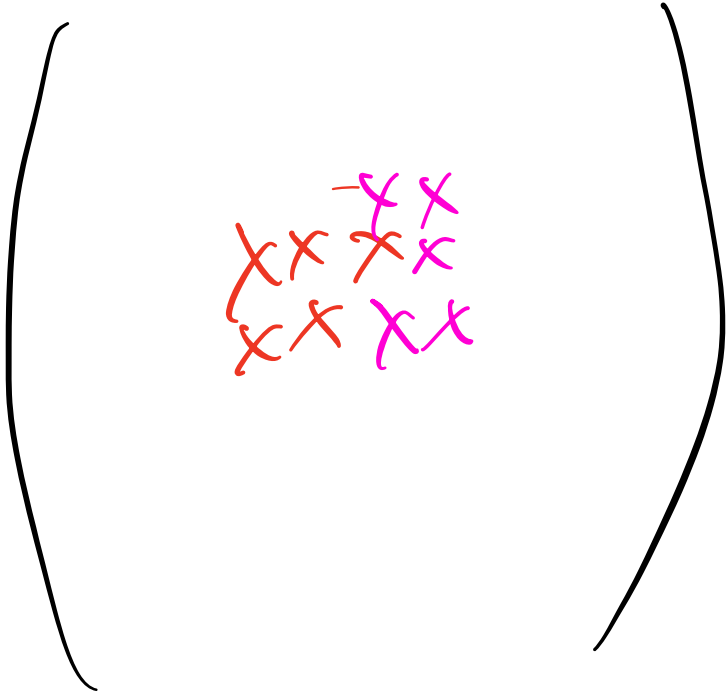
pixels
relevant
to model
 $f_\theta(z)$

Z value will help the model figure out what area to focus on

$P(x|z) = \text{Bernoulli}$ allows the model to learn, within the area of interest, which pixels to set 0 and which pixels to set to 1.

two colors

$$z = 0.1$$



x	x	x
x	x	x
x	x	x

$$f_{\theta}(z)$$

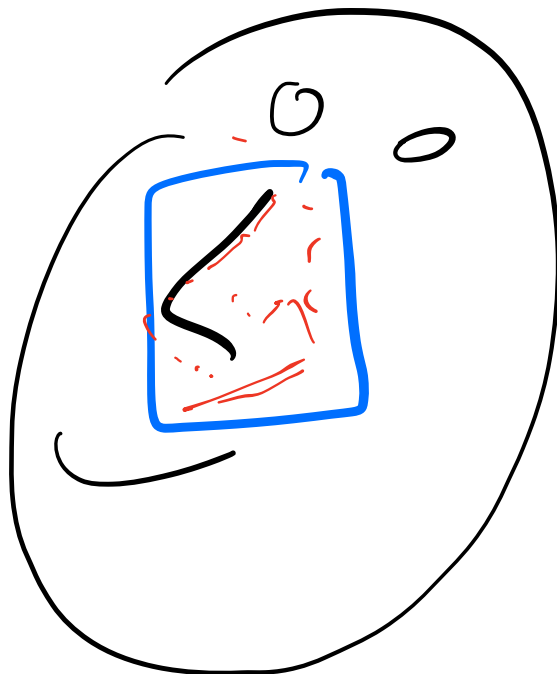
$$\begin{pmatrix} 0.1 & 0.1 & & 0.1 \\ & & & \\ & & & \\ 0.8 & & 0.1 & 0.2 \\ & & & \\ & & & \end{pmatrix}$$

$$Z \begin{pmatrix} 0.5 & 0.5 & & 0.5 \\ 0.5 & 0.5 & & 0.5 \\ & & \boxed{\begin{matrix} 0.1 & 0.01 \\ 0.9 & 0.98 \\ 0.95 & 0.91 \end{matrix}} & \\ 0.5 & & & 0.5 \end{pmatrix}$$

↗

1 → white

$\theta \rightarrow \text{block}$



$$\vec{z} \sim N(\vec{0}, I)$$

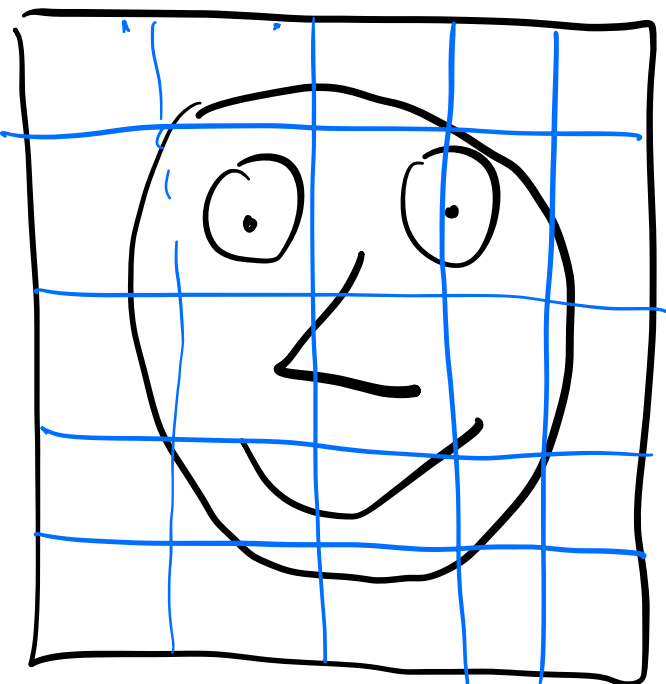
$$f_{\theta}(z)$$

training will adjust the

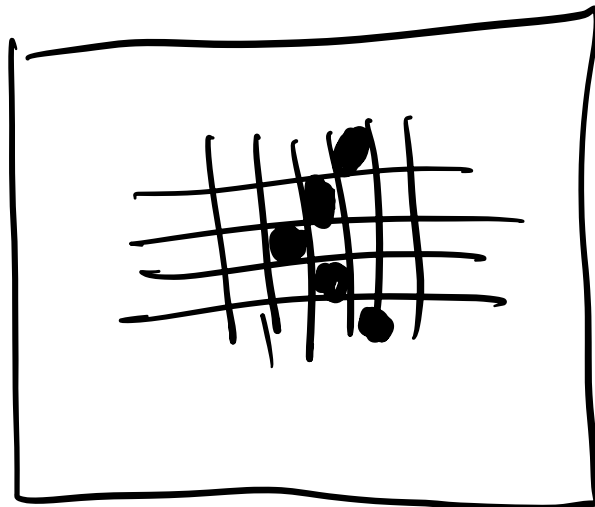
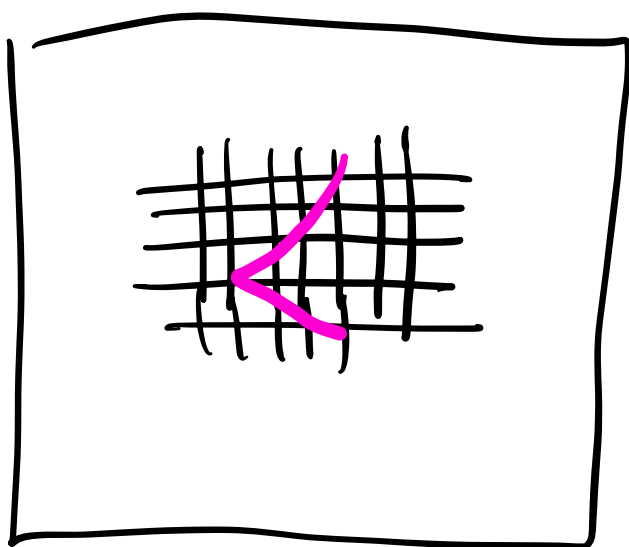
θ

$$\vec{z} \sim N(\vec{0}, I)$$

$$p(x|z) = \text{Bernoulli}(\underbrace{\phi(z)}_{z=0.1})$$

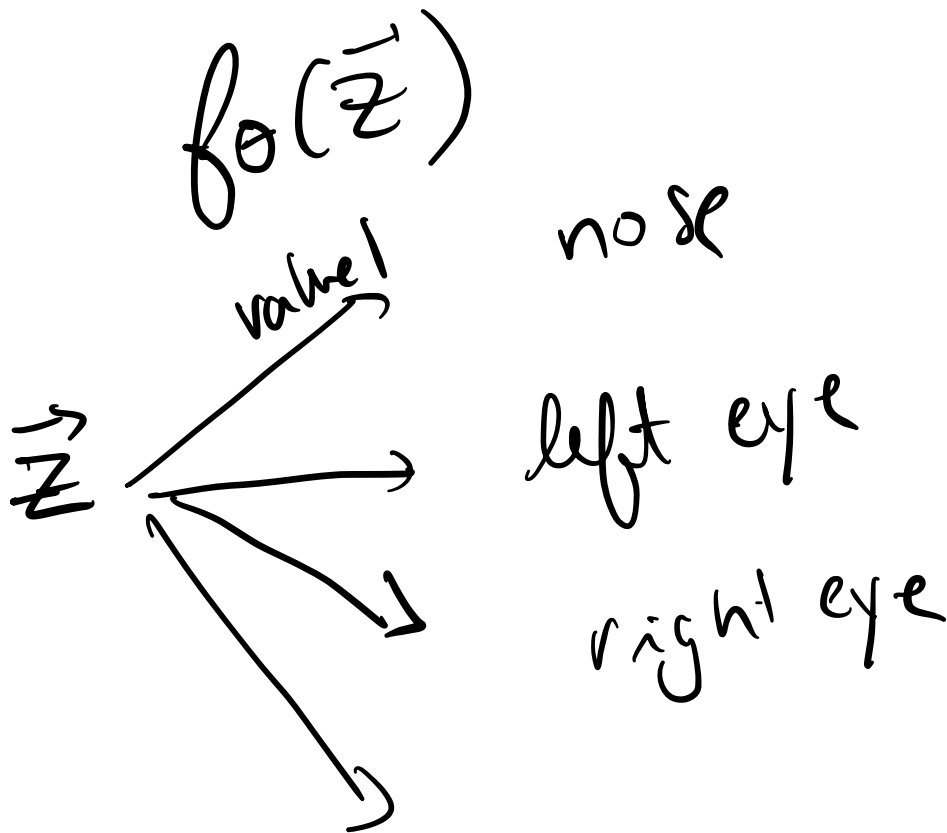


0.5	0.5	0.5	0.5	0.5
0.5	0.5	0.01	0.5	0.5
0.5	0.01	0.55	0.5	0.5
0.5	0.5	0.5	0.5	0.5
0.5	0.5	0.5	0.5	0.5



→
z →

$$\vec{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ z_4 \\ z_5 \end{bmatrix}$$



Let's approximate $p(\mathbf{z}|\mathbf{x})$

$$p(\mathbf{z}|\mathbf{x}) \approx \mathcal{N}(\mu_{\phi}, \Sigma_{\phi})$$

(red arrows pointing to μ_{ϕ} and Σ_{ϕ})

 "parameters"

choice in the model

learn these from

VAE

$$\begin{cases} p(z) = N(\vec{0}, I) \\ p(x|z) = \text{Bernoulli}(f_\theta(z)) \\ p(z|x) \approx N(\mu_\phi, \Sigma_\phi) \end{cases}$$

training the model according to

$$E_{p(z|x)} [\log p(x|z)] - \text{KL}(p(z|x), p(z))$$

not equal to $\log p(x)$ anymore

of latent variables?

Currently it's fixed but you will need to play with different numbers to figure out the optimal one.

HW3 released tomorrow morning